2025 #7

Ch3 미분 · 유형: 연쇄법칙

$f : \mathbb{R}^3 \to \mathbb{R}^2,\ g : \mathbb{R}^2 \to \mathbb{R}^3$ 가

$$f(u,v,w) = (u^3 + v^2 + 2u + 2v + w,\ u^7 + v^6 + w^8 + u + 2v - w)$$

$$g(x,y) = (x^5 + xy + y^3 + x + 2y,\ x^5y^7 + y^7 + 3x + 4y,\ x^{21} + y^{30} + 3x + 2y)$$

일 때 $D(f \circ g)(0,0)$ 을 구하라.

Step 1 **chain rule**: $D(f \circ g)(0,0) = Df\big(g(0,0)\big)\, Dg(0,0)$. 먼저 $g(0,0) = (0,0,0)$ 확인.

Step 1 $g(0,0) = (0,0,0)$ 이므로 Df 는 $(0,0,0)$ 에서 계산한다.

$$Df = \begin{pmatrix} 3u^2 + 2 & 2v + 2 & 1 \\ 7u^6 + 1 & 6v^5 + 2 & 8w^7 - 1 \end{pmatrix} \Rightarrow\ Df(0,0,0) = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 2 & -1 \end{pmatrix}$$

Step 2

$$Dg(0,0) = \begin{pmatrix} 5x^4 + y + 1 & x + 3y^2 + 2 \\ 5x^4y^7 + 3 & 7x^5y^6 + 7y^6 + 4 \\ 21x^{20} + 3 & 30y^{29} + 2 \end{pmatrix}_{(0,0)} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 3 & 2 \end{pmatrix}$$

Step 3

$$\begin{pmatrix} 2 & 2 & 1 \\ 1 & 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 2+6+3 & 4+8+2 \\ 1+6-3 & 2+8-2 \end{pmatrix}$$

Step 4 $$D(f \circ g)(0,0) = \begin{pmatrix} 11 & 14 \\ 4 & 8 \end{pmatrix}$$